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To cite this article: R Mitaphonna *et al* 2023 *J. Phys.: Conf. Ser.* **2582** 012033

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Identification possible preserved 2004 Indian Ocean tsunami deposit collected from Pulot Village in Aceh Besar Regency, Indonesia

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Abstract. Identification of the tsunami candidate layers deposited by the Indian Ocean tsunami in 2004 has been successfully performed in Pulot Village, located in Leupung district, Aceh Besar Regency, Aceh Province. Based on a study of tectonic activity before and after the earthquake, Pulot Village was reported as one of the potential locations for finding tsunami deposits. Tsunami sediment is collected by inserting a hand drill into the ground more than 1 m deep. The presence of the 2004 tsunami candidate layers was identified according to the Troels-Smith characteristics. Based on our visual analysis, the prospective tsunami layer is found at a depth of 4 cm with a thickness of 8-38 cm, which is characterized by fine-medium sized dark brown sand deposits on weathered paleo soil layers. The sharp boundaries with the layers above and below it and the presence of organic material (plant roots, stems and anthropogenic material) can also be clearly observed in the tsunami deposit layers. The candidate layer for the 2004 tsunami deposit found will be further studied geochemically using a laser-induced breakdown spectroscopy (LIBS) instrument.

Keyword: 2004 Indian Ocean tsunami; tsunami deposit; Pulot village; Aceh; sediment classification; megascopic analysis

1. Introduction

The massive Indian Ocean tsunami that happened at the end of 2004 was recorded as the deadliest natural disaster in modern history. This disaster has killed a quarter of a million people on the coasts of the Indian Ocean, destroyed infrastructure, damaged coastal systems and altered soil profiles and water quality [1]. Not only that, the tsunami waves have eroded the seabed and land so that tsunami sediments are widely deposited in inundation areas [2]. The severe damage and the large number of victims caused by this disaster indicate that the community's preparation and preparedness are still low due to the lack of knowledge and awareness of the community about the potential for an earthquake and tsunami [3-4]. Therefore, research on modern and prehistoric tsunamis that have hit is very important to be studied in depth to evaluate the vulnerability and readiness of coastal communities.

Deposits carried by tsunami waves contain important information about the tsunami event and it can be used to reconstruct the tsunami event [5]. However, unfortunately, the presence of tsunami deposits that are well preserved and protected from human activities is currently difficult to find. Pulot

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Village, which is located in the west of Aceh Province, became the target research area in this study because it has a flat morphology, is slightly sunken, protected from normal coastal currents and human activities, thus it is predicted that the tsunami residue will still be preserved in good condition. Additionally, Pulot Village is one of the potential locations for research on earthquakes and tsunamis based on coastal morphology and studies of soil uplift and subsidence due to tectonic activity before and after the earthquake. The lowest area between the coastal ridges of this area has developed into swamps that have the potential as a place for depositing modern tsunamis and prehistoric tsunamis so this place is suitable as a research location to learn more about tsunami events. The objective of this investigation is to find out if there are any tsunami deposits in various locations of Pulot Village and describe their properties by applying coring and paleo seismic techniques. In forthcoming research, we intend to use the LIBS method to perform geochemical analysis of the tsunami sediment samples. We have already employed the LIBS approach successfully in identifying chemical markers for soil samples that were impacted by the 2004 tsunami in Aceh Province, and we have discovered promising chemical markers that can distinguish between soils affected by tsunamis and those that are not [6].

2. Material and Methods

We took samples of tsunami sediment from two points in Pulot Village, Leupung District, a coastal area in the western part of Aceh Province that was hit by the devastating 2004 tsunami. Pulot Village has placed 32 km from the capital city of Banda Aceh and can be reached in approximately 30 minutes. Sampling points are located in swamp areas less than 2 km from the coastline as shown in Fig. 1.

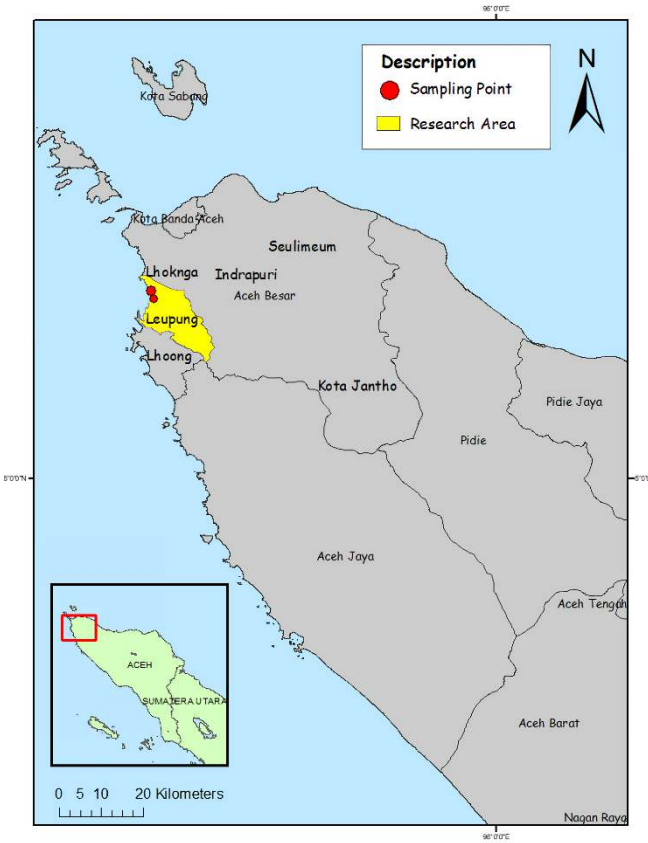

Figure 1. A map showing the location of the sample collection

Sampling begins with morphological observations around Pulot coastal area to find a suitable location for sampling tsunami sediment. The samples were taken by driving a hand drill into the ground until it reached an impenetrable part (up to 1 m). Description and identification of colour, visible grain size distribution, and further materials such as plant remains of roots, stems, and shells were also carried out

on the sediment samples taken in this study. Sediment samples that have been collected will be further processed in the laboratory including heating, grinding and sieving.

Figure 2. Drilling activities of tsunami deposit samples using coring in Pulot village b). Description of the sediment stratigraphic profile in Pulot village

3. Result and Discussions

In the vicinity of the Pulot coast, we carried out 2 core sediment drillings with a depth of more than 1 m, namely Pulot 1 dan Pulot 2. Furthermore, we also analyzed the grain size distribution, sedimentological features and interpretation of collected sediment as shown in Fig. 3. It could be demonstrated in Fig 3, the top part of investigated deposits consists of dark brown peat which has a thickness of 4-8 cm. In several parts, organic content such as plants and roots remains contained in these layers. This layer has uniform and well-sorted clay grain size which is suspected as new soil formed after the 2004 tsunami. Based on megascopic identification, the tsunami deposits usually are composed of very coarse gravel-sized material to lumps mixed with silt-clay-sized fine material. Furthermore, the tsunami deposit layer contains coral fragments, shells and plant fragments such as charcoal and roots. The tsunami deposit layer also has a clear boundary with the layers below and above it. Tsunami waves will come as a run-up by carrying sediment from the sea and depositing it on land, then backwash waves carry land sediments to the sea. This mechanism allows the accumulation of sedimentary material from land and sea so that it has a particle size distribution with poor sorting.

Based on our visual observations, sediment layers for the 2004 tsunami were identified at depths of more than 3 cm at each point. We found tsunami deposits at point 1 with a thickness of 12 cm, while at other points it was 38 cm. The tsunami deposit layers identified in this work have almost similar characteristics to tsunami deposits in other areas. We found a medium-fine, brown sand that was deposited on a paleo soil. The grain size is coarse at the bottom of the layer gradually becoming medium to fine at the top of the layer. There is a clear boundary with the layer below the tsunami deposit. In addition, in some areas, the candidate layer for the 2004 Indian Ocean tsunami contains fine roots, plant stems, and anthropogenic products (plastics). The top of the tsunami sand layer is usually a fine layer of silt or clay deposited when the tsunami waves reach a calm state before backwashing occurs. The top layer of tsunami sediment is usually silt or clay which was deposited when the tsunami waves reached a calm state before backwashing. The grain size of the tsunami deposit layer is also relatively different when compared to other layers. Based on previous research, the tsunami sediment layer has a relatively smaller grain size compared to the peat layers above and below it. The candidate tsunami layer has a bimodal and tri modal grain size distribution, rarely a unimodal distribution. However, the peat layer (especially the lower peat layer) is distributed unimodal and bimodal.

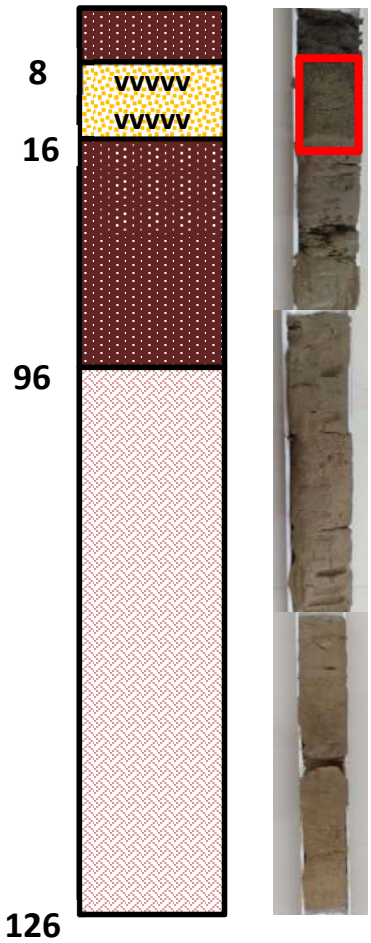

Figure 3. The stratigraphy profiles of deposit samples were collected from Pulot 1

In the eastern part of Simeulu Island, a tsunami deposit candidate was found at a depth of 22 cm and containing gravel to boulder sized material. Meanwhile, the prospective tsunami layer in Busong Simelue Bay was identified at a depth of 3–8 cm which is characterized by medium-coarse sand content, blackish brown, shell fragments and clear boundaries with the underlying layer [7]. Some of the tsunami deposits found on Lhoknga beach, Aceh Besar Regency, are characterized by fine-sized sand containing shell fragments of marine organisms in their layers [8]. The tsunami deposit collected from the coast of Jantang, Lhoong, Aceh Besar Regency, is fine to medium sized sand with very poor segregation [9]. The tsunami sediment layer found on the coast of Lhoong, Aceh Besar, contains animals with shells, rip-up clasts and sharp contacts with other layers [10]. Furthermore, the tsunami affected soil taken from three impacted regions in Aceh Province has non-uniform grain sizes which are dominated by coarse sand and fine silt [11].

Furthermore, a layer of blackish brown silt with a thickness of 80 cm and 32 cm is found under the tsunami candidate layer. This layer is sticky so it is very difficult to penetrate with a hand drill. In addition, the stratigraphy of Pulot 2 was found a layer of yellowish-brown sand beneath the clay layer at a depth of 76 cm. This layer has a thickness of 26 cm and grain size of fine sand from the top to the bottom of the layer. There are no remnants of plant roots or shell material in this layer. This sand layer is thought to be sediment carried by storm events because it has different characteristics from tsunami deposits, such as the absence of an upward smoothing trend, unimodal and bimodal distribution, and the absence of lamination at the bottom and top of this layer [12]. The grain size distribution carried by a hurricane event will inevitably be smaller than that carried by a tsunami event. This condition occurs

because the energy of the tsunami is very large compared to the energy of the storm. In some cases, tsunami waves can even carry Boulder-sized rocks (>25 µm in diameter) to a distance of several kilo meters from the shoreline. A hurricane can't carry a rock the size of a boulder, while a hurricane can lift as much sand and gravel.

Figure 4. The deposit samples were collected from Pulot 1 at a depth a). 0-35 cm b). 35-70 cm and c). 70-126 cm

4. Conclusions

In this present work, we have successfully identified the presence of 2004 Indian Ocean tsunami sediment layers taken from Pulot Village, which is one of the coastal areas that was stricken by the giant tsunami in 2004 in Aceh Province. The 2004 tsunami deposit layers at each point can be identified with a thickness of more than 5 cm. The tsunami deposit layer found in this research is fine-medium sized sand deposited on top of a layer of clay suspected to be paleo soil. Gradually, the finer grain size from the bottom of the layer to the top of the layer and a sharp boundary with the layer below can be clearly observed in the tsunami deposit layer. In general, the sediment layers for the 2004 Indian Ocean tsunami that we found in this study have characteristics that tend to be similar to those identified in other tsunami-affected areas.

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Acknowledgements
The authors are expressing appreciation for financial support for their research from the Directorate of Research and Community Services Directorate General of Research Strengthening and Development, Ministry of Research, Technology and Higher Education, Government of the Republic of Indonesia. This funding was received through the Higher Education Excellence Research Grants (Higher Education Excellence Basic Research, PDUPT) Year 2019, with the contract number 215/SP2H/LT/DPRM on March 8, 2019, as well as the Masters to Doctoral Education Research for Superior Bachelors (Masters Doctoral Undergraduate Program, PMDSU) in 2020, with the contract number 111/UN11.2.1/PT.01.03/DPRM/2020 dated August 4, 2020.